



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-15

Experiment 15. Feed Forward Neural Network

Aim

To implement a **Feed Forward Neural Network using Python.**

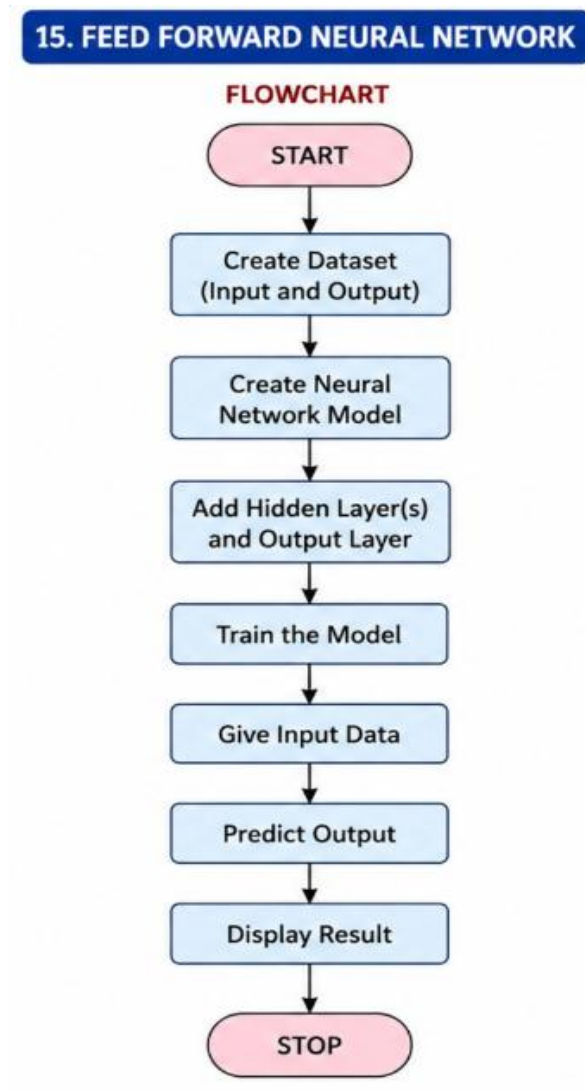
Objectives

- To understand the structure of a neural network.
- To create input and output layers.
- To train the network and make predictions.

Algorithm

1. Create the input dataset.
2. Create the neural network.
3. Add the hidden layer.
4. Train the model.
5. Predict the output.
6. Display the result.

Flowchart



Code

```
from  
  
sklearn.neural_network  
  
import MLPClassifier
```

```
X = [[0,0], [0,1], [1,0],  
[1,1]]
```

```
y = [0, 1, 1, 0]
```

```
model = MLPClassifier(  
  
    hidden_layer_sizes=(4,),  
  
    max_iter=2000,  
  
    random_state=1  
)
```

```
model.fit(X, y)
```

```
print("Predictions:",  
model.predict(X))
```

Output



The screenshot shows a Python IDE window titled 'ai lab 13.py - C:/Users/shrut/AppData/Local/Programs/Python/Python314/ai lab 13.py (3.14.2)'. The code in the editor is as follows:

```
from sklearn.neural_network import MLPClassifier  
  
X = [[0,0], [0,1], [1,0], [1,1]]  
y = [0, 1, 1, 0]  
  
model = MLPClassifier(  
    hidden_layer_sizes=(4,),  
    max_iter=2000,  
    random_state=1  
)  
  
model.fit(X, y)  
  
print("Predictions:", model.predict(X))
```

The output window on the right, titled 'IDLE Shell 3.14.2', shows the following output:

```
Python 3.14.2 (tags/v3.14.2  
on win32  
Enter "help" below or cli  
>>>  
= RESTART: C:/Users/shrut  
  
Warning (from warnings mc  
File "C:/Users/shrut\AppData/Local/Programs/Python/Python314/Scripts/python.exe"  
neural_network\multilayer_perceptron.py", line 111, in MLPClassifier  
warnings.warn(  
ConvergenceWarning: Stochastic gradient descent did not converge.  
mization hasn't converged yet.  
Predictions: [0 1 1 0]  
>>>
```

Result

Predictions: [0 1 1 0]

Conclusion

Thus, the Feed Forward Neural Network was successfully implemented using Python.